Lokesh Venkatesh

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Research Interests: **Complex Systems** | **Computational Neuroscience** | **Machine Learning**

Education

Indian Institute of Science Education and Research (IISER), Pune — Nov 2022 — Present
Integrated BS–MS | Natural Sciences — Interdisciplinary coursework across the basic sciences — CGPA: 7.7/10

Yuvabharathi Public School, Coimbatore — Jul 2022
AISSCE | Grade 12 (CBSE) — Physics, Mathematics, Chemistry, Informatics Practices, English — Grade: 92.6%

Yuvabharathi Public School, Coimbatore — Jul 2020
AISSE | Grade 10 (CBSE) — Science, Mathematics — Grade: 94.4%

Research Experience

Tumour-immune microenvironment interactions in cancer progression — Nov 2025 — Present
Advisors: Prof. Ram Rup Sarkar, NCL Pune & Dr. Chandrakala Meena, IISER Pune

- Carrying out fixed-time simulations of a 12-D ODE model with further parameter & bifurcation analyses to investigate how tumour-immune microenvironment interactions influence the dynamics of cancer progression.

Self-supervised Representation Learning for Circadian Time Series — May 2025 — Jul 2025
Advisor: Dr. Shaon Chakrabarti, NCBS Bangalore

- Implemented representation learning algorithms like the variational autoencoder architecture on patient actigraphy datasets, followed by clustering, in order to obtain their learned low-dimensional representations.
- Analysed the learnt embeddings to identify the respective circadian phases and isolate what specific variables carry the most significance in the multivariate time series.

Study of the Mitochondrial Genome using Motif Distributions — Aug 2024 — Apr 2025
Advisor: Prof. MS Madhusudhan, IISER Pune

- Analysed motif distributions across a database of mitochondrial genomes as a metric to both reconstruct phylogenetic relationships over the course of its evolution, as well as identify genomic translocations.
- Also looked at possible regulatory correlations between mitochondrial & nuclear genome promoter sequences.

Academic Projects & Training

Schools & Workshops

Spins, Games & Networks, 2025 — Dec 2025
Two-week fully-funded winter school on theoretical & computational approaches to complexity science at IMSc Chennai

NeuroAI — Jul 2025
Ten-day interactive online course on principles connecting artificial & biological neural nets by the Neuromatch Academy

Brain, Dynamics & Computation, 2025 — May 2025
Two-week fully-funded summer school on network science-approaches to computational neuroscience at IMSc Chennai

Course & Reading Projects

Tools for Complexity Science — Reading Project — Sep 2025 — Present
Undertaking a reading project (*Advisor: Dr. Chandrakala Meena*) to study the various methods & tools, largely based on Stefan Thurner et al's book, used to model complex biological and social systems' phenomena.

Relevant Coursework

Data Science — Applied Mathematical Methods, Deep Learning, Image & Video Processing, Mathematical & Statistical Optimisation, Computational & Functional Genomics, Signal Processing, Mathematical Finance

Physics — Classical Mechanics, Statistical Mechanics, Nonlinear Dynamics, Physics of Biological Systems

Mathematics — Multivariable Calculus, Linear Algebra, Probability, Graph Theory, Markov Processes

Biology — Systems Biology, Neurobiology, Bioinformatics, Cellular Biophysics I & II, Animal Behaviour, Ecology & Evolution, Cell Biology, Physiology, Genetics

Humanities & Social Sciences — Science & the World, Philosophy of Science, Cognitive Bases of Science, Science & Society, Public Policy I & II

Skills

Technical: High-Performance Computing, Self-supervised Representation Learning, Time Series Analysis, Bioinformatics Tools, Numerical Methods, Fixed-time Simulations

Programming: Python (Pandas, NumPy, PyTorch, Scikit-learn, Biopython, SciPy), Julia, Bash, Markdown, LaTeX

Languages: Bilingual proficiency in English & Tamil; Elementary proficiency in Hindi & Kannada

Scholarships & Awards

INSPIRE's Scholarship for Higher Education (SHE)
INSPIRE-SHE Scholar (4 Lakh INR $\approx 4,500$ USD) - awarded to top 0.1% (approx. 10,000) students nationwide by the Department of Science and Technology, Government of India

Extracurricular & Outreach

Academic Support Group, IISER Pune Aug 2023 — Sep 2024
Senior teaching & logistics volunteer for a student collective mentoring socio-economically underprivileged students

Kalpa — IISER Pune's Student Media House Mar 2023 — Jan 2024
Member of the student media body's core team that coordinated investigative journalism projects on campus matters

References

Prof. M.S. Madhusudhan (IISER Pune)
Email: madhusudhan@iiserpune.ac.in

Dr. Chandrakala Meena (IISER Pune)
Email: chandrakala@iiserpune.ac.in

Dr. Shaon Chakrabarti (NCBS Bangalore)
Email: shaon@ncbs.res.in